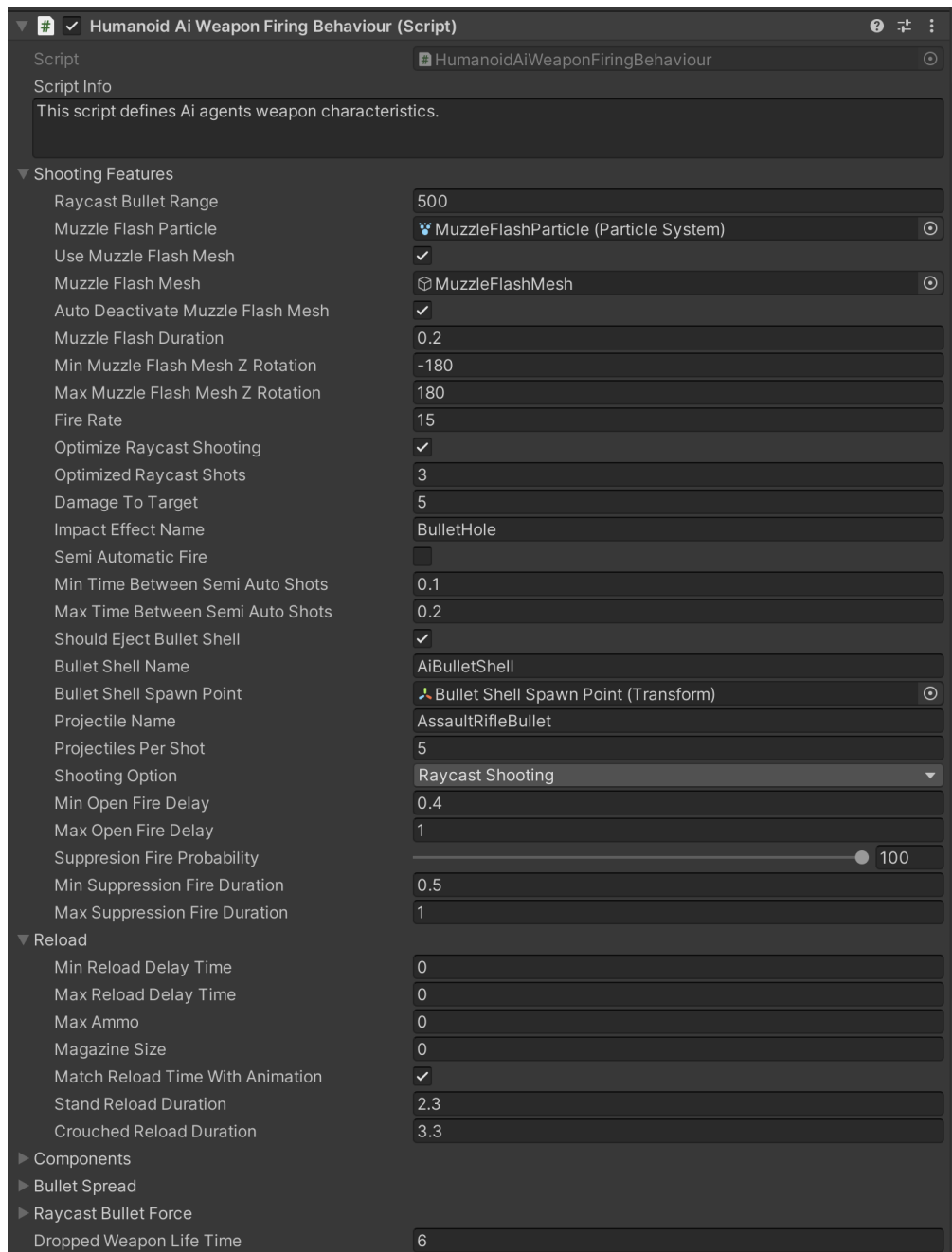


Weapon Firing Behaviour Part-1

Introduction

This is the Part-1 of Humanoid AI Weapon Firing Behaviour and in this part we have covered following topics - full automatic fire, Semi automatic fire, Reloading, Suppression fire, Dropped Weapon lifetime and basic setup required for the weapon to function properly.

Humanoid AI Weapon Firing Behaviour



Fields	Info
Raycast Bullet Range	This field is applicable to raycast and projectile shooting determining how far this humanoid AI agent weapon can shoot.
Muzzle Flash Particle	Drag and drop Muzzleflash particles from the hierarchy (child of the Shooting Point gameobject attached to AI agent`s weapon).
Use Muzzle Flash Mesh	Use mesh instead of particles for muzzle flash effect.

Muzzle Flash Mesh	Drag and drop MuzzleFlashMesh gameobject from the hierarchy (child of the Shooting Point gameobject attached to AI agent`s weapon).
Auto Deactivate Muzzle Flash Mesh	Deactivates 'MuzzleFlashMesh' right after weapon shot animation clip playback. Otherwise it will get deactivated after set number of seconds specified in the 'TimeToDeactivateMuzzleMesh' field.
Muzzle Flash Duration	Muzzle Flash mesh deactivation delay.
Min Muzzle Flash Mesh Z Rotation	Set the minimal rotational offset angle value for MuzzleFlashMesh to add variations of its appearance with each new shot.
Max Muzzle Flash Mesh Z Rotation	Set the maximal rotational offset angle value for MuzzleFlashMesh to add variations of its appearance with each new shot.
Fire Rate	Shots per second parameter of this weapon.
Optimize Raycast Shooting	If checked, will decrease performance hit by doing raycast shot only once for the first shot at the current closest enemy and after that performing set number of consecutive shots inside field below named 'OptimizedRaycastShots' without actual raycasting until target is killed or another enemy becomes closest target in which case raycast optimization will reset itself for that new closest enemy.
Optimized Raycast Shots	Specifies how many consecutive shots after first raycast shot (not including it) AI agent will do at its current target before resuming OptimizeRaycastShooting cycle.
Damage To Target	Amount of damage points of each raycast shot of this weapon (Does not affect projectiles).
Impact Effect Name	Enter the exact name of the default impact effect from the object pooler that will be applied to surfaces which do not have their own impact effects specified in 'HitImpactEffect' script attached to them.

Semi Automatic Fire	If checked, then this weapon will only do single shots and would not be able to do burst (automatic) firing.
Min Time Between Semi Auto Shots	Minimal time interval between semi-automatic shots.
Max Time Between Semi Auto Shots	Maximal time interval between semi-automatic shots.
Should Eject Bullet Shell	If enabled, then bullet shells will be ejected when firing this weapon.
Bullet Shell Name	Name of the bullet shell effect inside object pooler.
Bullet Shell Spawn Point	Drag and drop child gameobject of this weapon named 'BulletShellSpawnPoint' from the hierarchy into this field. Bullet shells particle prefabs will then be spawned at the origin of this gameobject and will along the local Z axis of this game object.
Projectile Name	Copy and paste name of the projectile to be used with this weapon from the 'ObjectPooler' script into this field.
Projectiles Per Shot	Enter the number of the projectiles of the single shot of this weapon if you want this weapon to be shotgun. The spread of those projectiles is specified in 'BulletSpread' paragraph.
Shooting Option	Select one of the three available options for the kind of shooting that this weapon will utilise. RaycastShootingWithTracers is an option where those tracers are not dealing damage to targets and are there only for visuals.
Min Open Fire Delay	Minimum time delay before commencement of the weapon firing behavior.
Max Open Fire Delay	Maximum time delay before commencement of the weapon firing behavior.

Suppression Fire Probability	Sets the probability for AI agent to continue firing his weapon (for some time specified in the fields below) at the targets that became hidden behind covers.
Min Suppression Fire Duration	Specify the minimum duration of the suppressing fire.
Max Suppression Fire Duration	Specify the maximum duration of the suppressing fire.
Min Reload Delay Time	Minimal delay time before reloading the weapon after shooting last bullet from magazine or in cases that require manual reload after each single shot.
Max Reload Delay Time	Maximal delay time before reloading the weapon after shooting last bullet from magazine or in cases that require manual reload after each single shot.
Max Ammo	Maximum amount of rounds for this weapon that AI agent will have.
Magazine Size	Amount of rounds in the magazines for this weapon.
Match Reload Time With Animation	If checked then duration of the reload will match the duration of the used reload animation clip. If unchecked then it will be possible to set the durations of reload for Crouched and Standing positions regardless of the reload animation clip playback time.
Stand Reload Duration	The time it will take to reload this weapon in stance position.
Crouched Reload Duration	The time it will take to reload this weapon in crouched position.
Weapon Mesh	Drag&Drop this weapon mesh from this AI agent's hierarchy into this field.
	Drag and drop 'CoreAiBehaviourScript' component attached with the root gameobject of this AI agent from the hierarchy into this field.

Core Ai Behaviour Script	
Animator Component	Drag and drop 'Animator' component attached with the root gameobject of this Ai agent from the hierarchy into this field.
Weapon Sounds	Drag&drop 'WeaponSounds' gameobject with 'AlertingSoundActivator' script attached to it from this AI agent's hierarchy into this field.
Stand Reload Clip	Drag and drop the stand reload animation clip from the project window into this field.
Crouch Reload Clip	Drag and drop the crouch reload animation clip from the project window into this field.
Stand Fire Clip	Drag and drop the stand fire animation clip from the project window into this field.
Crouch Fire Clip	Drag and drop the crouch fire animation clip from the project window into this field.
Dropped Weapon Life Time	Specify for how long this weapon will stay active after Ai agent that carried it was killed.